

Model Paper - Database Management Systems

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Q1 (20 marks)

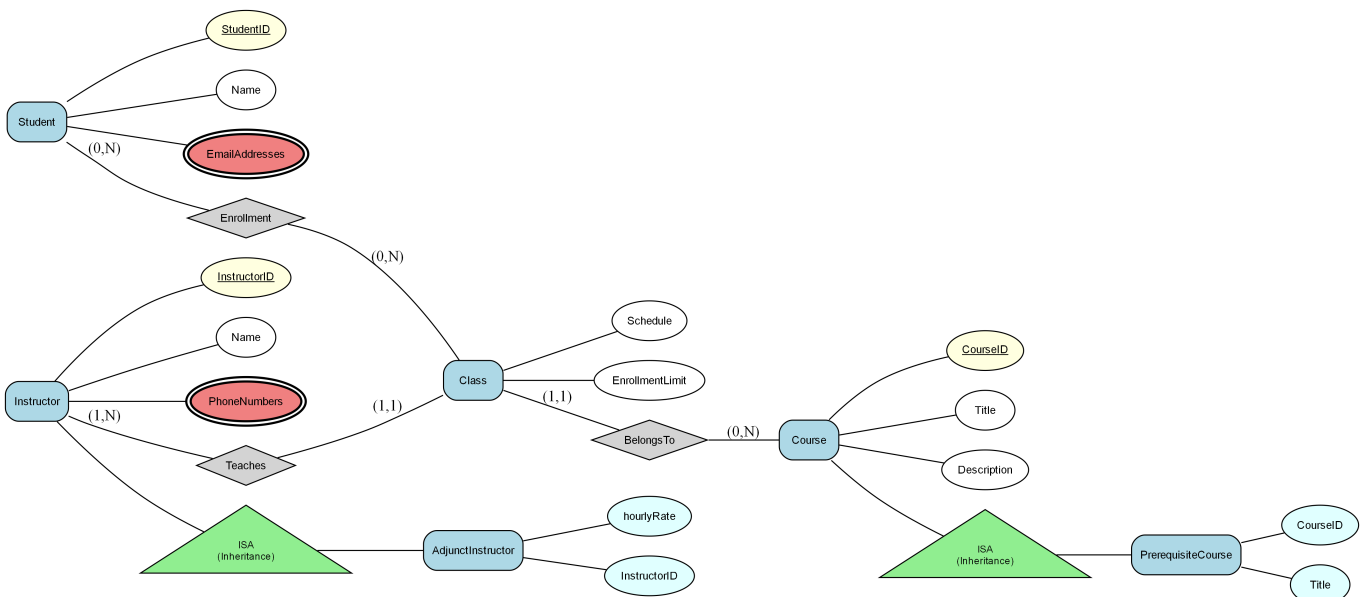
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A university is managing its classes, students, instructors, and courses. The 'Student' entity contains attributes such as StudentID, Name, and EmailAddresses (a multivalued attribute). The 'Instructor' entity includes InstructorID, Name, and PhoneNumbers (another multivalued attribute). The 'Course' entity has CourseID, Title, and Description as its attributes. The 'Class' entity represents a relationship between 'Course' and 'Instructor', with attributes such as Schedule and EnrollmentLimit. The relationship between 'Student' and 'Class' is many-to-many, indicated by an Enrollment table. The 'Instructor' has a specialization subtype called 'AdjunctInstructor' with an attribute for hourlyRate, while the 'Course' has a prerequisite subtype called 'PrerequisiteCourse'. A Instructor can be related to many Classs through the 'Teaches' relationship. A Student can be related to many Classs, and a Class can be related to many Students through the 'Enrollment' relationship (Student participation is optional, Class participation is optional). Many Classs can be related to one Course through the 'BelongsTo' relationship (Course participation is optional). Draw the EER diagram for this scenario and include proper cardinalities

Figure: EER



- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R :
 $F = \{AB \rightarrow C, C \rightarrow D, D \rightarrow E, A \rightarrow B\}$. Analyze the schema and its dependencies to assess its normalization.

- a) Find all the keys in relation R using the attribute closure method. (4 marks)
- b) Is R in 3NF? Give reasons for your conclusion. (3 marks)
- c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A university database system is designed to manage student enrollment, course registrations, and faculty records. The database schema has been defined and includes relations such as `Students(sid: string, name: string, major: string, gpa: real)`, `Courses(course_id: string, title: string, credits: integer)`, and `Enrollments(sid: string, course_id: string, semester: string)`. The database system needs to ensure data integrity and proper management of course offerings each semester.

- a) Write SQL queries to in type 2 driver, jdbc api calls are converted to native java api calls. accept or refute the above statement justifying your answer. (2 marks)
- b) Code Segment:

```
``java String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql); pstmt.setString(1, "IT"); ResultSet rs =
pstmt.executeQuery(); ``
```

 Which type of JDBC statement is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)
- c) Briefly explain the options available for transferring data between two sources such as between tables and servers. (2 marks)
- d) Briefly explain how authentication and authorization are achieved in SQL Server. (3 marks)
- e) A university is planning to implement a new student management system. Describe the roles and responsibilities of different Students involved in database management, including a senior database administrator, junior database administrators, and data entry operators. Emily is a junior database administrator managing specific databases within the system.. Michael is a data entry operator responsible for inputting data into the system.. Nathan is a database developer responsible for designing and implementing database schemas.. Sarah is the senior database administrator tasked with overseeing the entire database system's creation and maintenance..**
- i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)
- ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)
- iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing

the Enrollments database using a user-defined server role. (5 marks)

iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)

v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the Students and Enrollments databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for a university: Student (studentId: int, name: varchar(100), email: varchar(50), phone: varchar(15)) Course (courseId: int, title: varchar(100), credits: int, department: varchar(50)) Enrollment (enrollmentId: int, studentId: int, courseId: int, grade: char(1)) The 'Student' table stores information about students enrolled in the university, including their unique ID, name, email address, and phone number. The 'Course' table includes details on courses offered, such as a unique course ID, title, number of credits, and department offering the course. The 'Enrollment' table manages the relationship between students and courses, storing each enrollment's unique ID, associated student ID, course ID, and the grade assigned to the student for that course. The 'Course' table holds details about courses, such as unique course ID, course name, credits, and department. Fee (feeId: int, studentId: int, amount: real, paymentStatus: varchar(20)) The 'Fee' table stores fee/payment details including payable amount and payment status.

a) Write SQL Queries to perform the following: i. Find the names and emails of students enrolled in a course titled 'Database Systems'.

- i. Find the names and emails of students enrolled in a course titled 'Database Systems'. (4 marks)
- ii. Find the student with the highest aggregate value using GROUP BY on related enrollment records. (6 marks)
- iii. Find the title and name from Course, Student, and Enrollment where Enrollment.enrollmentId is not null. (7 marks)

b) b) Create a function to calculate the total fee amount for a given student. This function should account for fee with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) c) Create a trigger that automatically updates the 'amount' column in the 'Fee' table whenever a new fee record is added, or an existing fee record is updated. The 'amount' column should contain the total fee amount for each student. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by student. The student pays only 50% of the total fee amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)